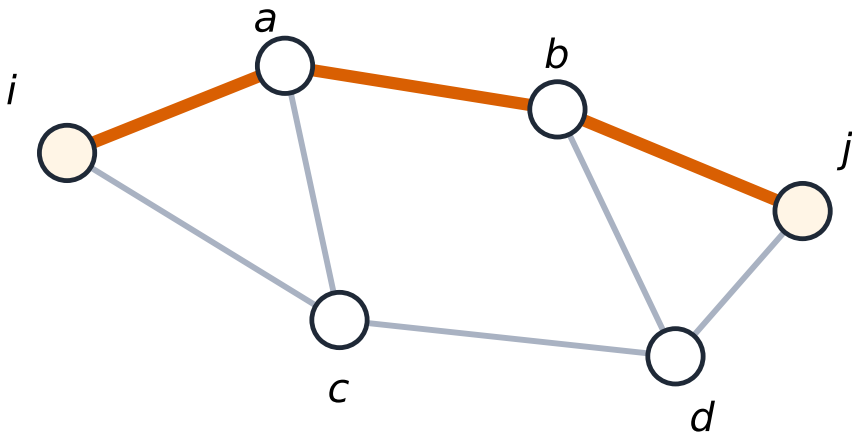


(a) Input graph geodesic

$\delta_{ij}^{(k)}$ on the fixed k -NN graph G_k

edge weights are measured in the input space

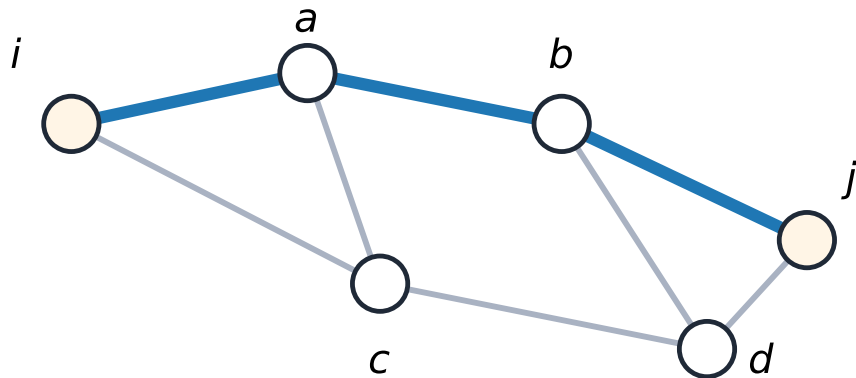


$$\delta_{ij}^{(k)} = w_{ia} + w_{ab} + w_{bj}$$

(b) Embedded input geodesic

$d_k^{\text{emb}}(i, j; \mathbf{Z})$ along $\gamma_{ij}^{(k)}$

same route as in (a), but edge lengths come from the embedding



route fixed by the input geometry

$$d_k^{\text{emb}}(i, j; \mathbf{Z}) = \|z_i - z_a\| + \|z_a - z_b\| + \|z_b - z_j\|$$